



The 66th ASH Annual Meeting Abstracts

POSTER ABSTRACTS

722.ALLOGENEIC TRANSPLANTATION: ACUTE AND CHRONIC GVHD AND IMMUNE RECONSTITUTION

Fecal Microbiota Transplantation to Prevent Acute Graft-Versus-Host Disease: First Pre-Planned Interim Analysis of a Randomized Trial

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In a randomized trial of fecal microbiota transplantation (FMT) vs. placebo after allogeneic hematopoietic cell transplantation (alloHCT), we demonstrated safety and feasibility of this approach and its powerful microbiota restorative potential (PMID: 37235836). The dose was small (5 capsules) and product from 4 donors was used without pooling. Microbiota engraftment (i.e. fraction of post-FMT microbiota coming uniquely from the donor) was 25-30%. Although acute GVHD (aGVHD) occurred numerically more frequently after FMT, post-hoc analysis showed a strong negative association between microbiota engraftment and aGVHD, as well as a donor effect (PMID: 37787998). These findings raised the hypothesis that we may reduce aGVHD rates by 2 strategies: (i) dose escalation and (ii) optimization of donor choice.

On November 1, 2023, we initiated the single-arm run-in phase of a randomized phase 2 double-blind trial of oral encapsulated third-party FMT vs. matched placebo (IND # 29935, NCT06026371) at Fred Hutch Cancer Center. The run-in phase has now been completed, with 6 months of follow-up. Each patient received FMT from 1 of 3 donors, with the primary endpoint of finding the donor with the highest mean microbiota engraftment. This donor would be used in the randomized phase, where patients are randomized (stratified for GVHD prophylaxis and donor type) between FMT vs. placebo, with the primary endpoint of grade III-IV aGVHD. Key inclusion criteria (both phases): age ≥ 18 , GVHD prophylaxis using a calcineurin inhibitor (CNI) + MTX or MMF, CNI + MMF + sirolimus, or CNI + PTCy (with or without MMF). Key exclusion criterion: mismatched (<9/10) unrelated donor. Patients received 3 capsules/day for 7 days, starting at neutrophil recovery and 2 days after discontinuation of antibacterial antibiotics. Each capsule contained $1-2 \times 10^{11}$ bacteria with >40% viability, with total dose per patient >4-fold higher than the prior trial. Stool samples were obtained at baseline (before transplant conditioning), pre-FMT, and 4 weeks post-FMT. Microbiota was profiled using (i) 16S rRNA gene sequencing (V3-V4) and DADA2-based exact amplicon sequence variants to permit comparison with the previous trial and (ii) shotgun metagenomic sequencing targeting 20M paired reads per sample, with Metaphlan4 for species-level assignment and sameStr for strain-level engraftment analysis. Here we report the results from the first pre-planned interim analysis at the end of the run-in phase.

20 patients were treated and followed for clinical endpoints; 17 of these patients provided samples for engraftment analysis. No grade 3+ treatment-related AEs occurred. The only DLT was an *H. pylori* infection. Although the FMT donor for this patient had a positive stool antigen, deep shotgun sequencing of their stool failed to identify *H. pylori* DNA, making infection transmission unlikely. Day +180 NRM was 0%. Grade III-IV aGVHD (2 acute, isolated stage 3 skin; 1 late-acute, isolated stage 3 lower gut) occurred in 3 patients, all of whom had received FMT from donor 1. Marked loss of diversity and compositional shift between baseline and pre-FMT was followed by a significant rise in diversity to levels close to normal levels and compositional shift toward donor post-FMT, all better for donors 2 and 3 than 1. In 16S-based engraftment analysis, median engraftment was 42% (33%, 47%, and 47% for donors 1, 2, and 3, respectively), ~15% higher than in the previous trial. Strain-level analysis showed 48%, 53%, and 50% mean engraftment for these donors, respectively. However, 2 patients treated with donor 2 engrafted poorly (9% and 16% engraftment). Short-chain fatty acid- and indole-producing, commensal obligate anaerobes (e.g. *Dorea longicatena*, *Alistipes putredinis*, *Ruminococcus bromii*, and *Eubacterium rectale*) and *Bifidobacterium longum* showed 100% donor engraftment. With equivalent results for donors 2 and 3 (both of whom yielded better microbiota and

clinical outcomes than donor 1) and considering our only DLT and less uniform engraftment with donor 2, we selected donor 3 as the optimal donor.

In summary, the run-in phase met its primary endpoint: improved microbiota engraftment through dose escalation and donor selection. The randomized phase opened to enrollment in June 2024, using FMT from donor 3 vs. placebo. Precision design of FMT trials with dose and donor optimization is feasible, with potential to improve microbiota and clinical outcomes.

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